Assignment
Digital Electronics

Q1. convert :
(i)
(a) $(85.63)_{10} \rightarrow (X)_2$

2	85	
2	42	1
2	21	0
2	10	1
2	5	0
2	2	1
2	1	0
	0	1

$(85)_{10} = (1010101)_2$

now, 0.63 decimal

$0.63 \times 2 = 1.26$ 1
$0.26 \times 2 = 0.52$ 0
$0.52 \times 2 = 1.04$ 1
$0.04 \times 2 = 0.08$ 0
$0.08 \times 2 = 0.16$ 0
$0.16 \times 2 = 0.32$ 0
$0.32 \times 2 = 0.64$ 0
$0.64 \times 2 = 1.28$ 1

$(0.63)_{10} = (.10100001)_2$ (approx

hence $(85.63)_{10} = (1010101.10100001)_2$

$X = 1010101.10100001$

(b) $(300.45)_{10} = (X)_8$

8	300	
8	37	4
8	4	5
	0	4

$(300)_{10} = (454)_8$

now. 45 decimal

$.45 \times 8 = 3.60$	3
$.60 \times 8 = 4.80$	4
$.80 \times 8 = 6.40$	6
$.40 \times 8 = 3.20$	3
$.20 \times 8 = 1.60$	1

$\Rightarrow (.45)_{10} = (.34631..)_8$

hence $(300.45)_{10} = (454.34631..)_8$

$\Rightarrow X = 454.34631..$

(c) $(F3A7C2)_{16} = (X)_2$

In hexadecimal

$F = 1111$

$3 = 0011$

$A = 1010$

$7 = 0111$

$C = 1100$

$2 = 0010$

Hexadecimal:-

0	0000	A	1010
1	0001	B	1011
2	0010	C	1100
3	0011	D	1101
4	0100	E	1110
5	0101	F	1111
6	0110		
7	0111		
8	1000		
9	1001		

$\Rightarrow (F3A7C2)_{16} = (1111\ 0011\ 1010\ 0111\ 1100\ 0010)_2$

(d) $(1101.101)_2 = (x)_{10}$

First $(1101)_2 = (1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0)_{10}$
$= (8+4+1)_{10} = (13)_{10}$

then $(.101) = (1\times2^{-1} + 0\times2^{-2} + 1\times2^{-3})_{10}$
$= (0.5+0+0.125) = (0.625)_{10}$

$\Rightarrow (1101.101) = (13.625)_{10}$

$\Rightarrow X = 13.625$

(e) $(1101010)_2 = (X)_{gray}$

gray code: First bit same, then each bit = XOR of previous and current bit

Binary = 1 1 0 1 0 1 0

First bit = 1

2nd $1 \oplus 1 = 0$

3rd $0 \oplus 1 = 1$

$1 \oplus 0 = 1$

$0 \oplus 1 = 1$

$1 \oplus 0 = 1$

$0 \oplus 1 = 1$

$\Rightarrow$ gray code of $(1101010)_2 = (1011111)_{gray}$

$\Rightarrow X = 1011111$

(ii)(a) 27

2	27	
2	13	1
2	6	1
2	3	0
2	1	$\cancel{0}$
	0	1

$\rightarrow$ Binary $= (1\cancel{0}011)_2$

8 bit form $= \boxed{(00010011)_2}$

$\rightarrow$ ASCII

$\boxed{\begin{array}{l} 2 = 50 \text{ in ASCII} \\ 7 = 55 \text{ in ASCII} \end{array}}$

Binary: $1\cancel{0}011$

First bit = 1

2nd $= 1 \oplus \cancel{0} = 0$

3rd $0 \oplus \cancel{0} = 0$

4 $1 \oplus 0 = 1$

5 $1 \oplus 1 = 0$

$\boxed{\text{gray} = 10010}$

Excess 3

$2+3=5 = (0101)_2$

$7+3=10 = (1010)_2$

Excess-3 code: (0101 1010)

b) 396

→ Binary

2	396	
2	198	0
2	99	0
2	49	1
2	24	1
2	12	0
2	6	0
2	3	0
2	1	1
	0	1

Binary = $(110001100)_2$

→ ASCII

3 = 51 in ASCII
9 = 57 in ASCII
6 = 54 in ASCII

→ gray code

Binary :- 110001100

1st bit = 1
2 = $1 \oplus 1 = 0$
3 = $0 \oplus 1 = 1$
4 = $0 \oplus 0 = 0$
5 = $0 \oplus 0 = 0$
6 = $1 \oplus 0 = 1$
7 = $1 \oplus 1 = 0$
8 = $0 \oplus 1 = 1$
9 = $0 \oplus 0$ 0

gray code : 101001010

→ Excess 3

$3+3=6 = (0110)_2$
$9+3=12 = (1100)_2$
$6+3=9 = (1001)_2$

⇒ Excess 3 = (0110 1100 1001)

(C) 4096

→ Binary

2	4096	
2	2048	0
2	1024	0
2	512	0
2	256	0
2	128	0
2	64	0
2	32	0
2	16	0
2	8	0
2	4	0
2	2	0
2	1	0
	0	1

Binary = $(1000000000000)_2$

→ Excess 3 code

$4+3 = 7 = (0111)$
$0+3 = 3 = (0011)$
$9+3 = 12 = (1100)$
$6+3 = 9 = (1001)$

⇒ (0111 0011 1100 1001)

→ ASCII

4 = 52 in ASCII
0 = 48
9 = 57
6 = 54

→ gray 1 0000 0000 0000

↳ 1 1 00 0000 00000

(iii) $3^2 = 9$

simple procedure to convert base 3 to base 9.

Procedure:

1. Pair the number is groups of 2 since $3^{2} = 9$ from right side
2. For each pair calculate 9 representation as
   3x (1st element) + 2nd element
3. Write the 9 representations in same order.

Applying to $(2110201102220112)_3$

1. grouping: 21 10 20 11 02 22 01 12

value = 3(2)+1 3(1)+0 3(2) 3(1)+1 3(0)+2 3(2)+2 +1 3(1)+2

7 3 6 4 2 8 1 5

Base 9 number $= (73642815)_9$

(iv) Binary arithmatic operations :-

a) +42 + (−13)

2	42	
2	21	0
2	10	1
2	5	0
2	2	1
2	1	0
	0	1

2	13	
2	6	1
2	3	0
2	1	1
	0	1

$(42)_{10} = (101010)_2$

$= (00101010)_2$

$(13)_{10} = (1101)_2$

$= (00001101)_2$

since -13 is -ve, 1's compliment : 11110010
2's compliment : $\frac{\begin{array}{r}+1\end{array}}{11110011}$

addition: 00101010
11110011
1 00011101
discarded

Result : $(00011101)_2$

b) $(-42) - (-13)$
$(-42) + (+13)$

$(13)_{10} = (00001101)_2$

$(42)_{10} = (00101010)_2$

since -42 is -ve : 1's compliment : 11010101
2's compliment : $\frac{\begin{array}{r}+1\end{array}}{11010110}$

addition: 11010110
+ 00001101
$(11100011)_2$

Result : $(11100011)_2$

Q2. (a) system has two identical circuits in parallel
say we have X and Y outputs
when circuits are good $X = Y$
when error, $X \neq Y$ (sometimes)

failure detection method :
Feed the outputs to an X-OR gate
say $F = X \oplus Y$
when $X = Y \Rightarrow F = 0$ (normal)
but when $X \neq Y \Rightarrow F = 1$ (error detected).

b) Boolean redutions

(i) $A'C' + A \ast B \ast C + AC'$

$= C'(A'+A) + A \ast BC$
$= C' + ABC$

Indentity $X + X'Y = X+Y$

$\Rightarrow C' + A \ast BC = C' + A(B)(C) = C' + AB$

$= \underline{C' + AB}$ $\underline{\text{Three litirals.}}$

(ii) $A'B(D' + C'D) + B(A + A'CD)$

here $D' + C'D = D' + C'$ using Idenity $X+X'Y = X+Y$
also $A + A'CD = A + CD$

$\Rightarrow$ exp $= A'B(D'+C') + B(A + CD)$
$= B(A'D' + A'C' + A + CD)$
$= B(A + A'C' + A'D' + CD)$
$= B(A + C' + D' + CD)$ using same Identity

$= B(A + C' + CD + D')$
$= B(A + C' + D + D')$
$= B((A+D') + (C+D'))$
$= B(A + C' + 1)$
$= B(1)$
$= \underline{B}$ $\underline{\text{one Literal}}$

(iii) $(A'+C)(A'+C')(A+B+C'D)$

$\cancel{(A'A' + A'C' + CA' + CC')(A+B+C'D)}$

$= \cancel{A'(A'C + CC')(A+B+C'D)}$

$= \cancel{A'(A'C)(A+B+C'D)}$

using law $(X+Y)(X+Z) = X+YZ$

$= (A' + CC')(A+B+C'D)$

$= A'(A+B+C'D)$

$= A'A + A'B + AC'D$

$= \underline{A'B + AC'D}$ 4 literals

Q3. Convert of standard SOP and POS forms

(i) $(AB+C)(B+C'D)$

$F = AB(B+C'D) + C(B+C'D)$

$= ABB + ABC'D + CB + CC'D$

here $ABB = AB$
$CC'D = 0$

$\Rightarrow F = AB + ABC'D + CB$

$= AB + BC + ABC'D$

since $X + XY = X$

$\Rightarrow AB + ABC'D = AB$

$\Rightarrow$ $\underline{F = AB+BC}$

Canonical SOP (minterms)

and POS (maxterms

A	B	C	D	AB	BC	F	Minterm / Maxterm	
0	0	0	0	0	0	0		M_0
0	0	0	1	0	0	0		M_1
0	0	1	0	0	0	0		M_2
0	0	1	1	0	0	0		M_3
0	1	0	0	0	0	0		M_4
0	1	0	1	0	0	0		M_5
0	1	1	0	0	1	1	m_6	
0	1	1	1	0	1	1	m_7	
1	0	0	0	0	0	0		M_8
1	0	0	1	0	0	0		M_9
1	0	1	0	0	0	0		M_{10}
1	0	1	1	0	0	0		M_{11}
1	1	0	0	1	0	1	m_{12}	
1	1	0	1	1	0	1	m_{13}	
1	1	1	0	1	1	1	m_{14}	
1	1	1	1	1	1	1	m_{15}	

from $F=1$ SOP $= \sum m(6,7,12,13,14,15)$

$F=0$ POS $= \prod M(0,1,2,3,4,5,8,9,10,11)$

(ii) $= X' + X(X+Y')(Y+Z')$

$F = X' + X(1)(Y+Z')$

$= X' + X(Y+Z')$

$= X' + XY + XZ'$

$= \underline{X' + XZ'} + XY$

$= X' + Z' + XY$ (as $X + X'Y = X+Y$)

$= X' + XY + Z'$

$= X' + Y + Z'$ (as $X + X'Y = X+Y$)

X	Y	Z	X'	Y	Z'	F=X'+Y+Z'	Minterm	Maxterm
0	0	0	1	0	1	1	m_0	-
0	0	1	1	0	0	1	m_1	-
0	1	0	1	1	1	1	m_2	-
0	1	1	1	1	0	1	m_3	-
1	0	0	0	0	1	1	m_4	-
1	0	1	0	0	0	0	-	M_5
1	1	0	0	1	1	1	m_6	-
1	1	1	0	1	0	1	m_7	-

∴ Canonical SOP = $\sum m(0,1,2,3,4,6,7)$

POS = $\Pi M(5)$

(iii) $B'D + A'D + BD$

$D(B' + A' + B)$

$D(A' + (B+B'))$

$D(A'+1)$ (as $B+B'=1$)

$D(1) = D$

A	B	D	B'	A'	B'D	A'D	BD	F	Min	Max
0	0	0	1	1	0	0	0	0		M_0
0	0	1	1	1	1	1	0	1	m_1	
0	1	0	0	1	0	0	0	0		M_2
0	1	1	0	1	0	1	1	1	m_3	
1	0	0	1	0	0	0	0	0		M_4
1	0	1	1	0	1	0	0	1	m_5	
1	1	0	0	0	0	0	0	0		M_6
1	1	1	0	0	0	0	1	1	m_7	

∴ Conical SOP = $\sum m(1,3,5,7)$

POS = $\Pi M(0,2,4,6)$

(iv) $(xy+z)(xz+y)$

$xy(xz+y) + z(xz+y)$

$xyxz + xyy + zxz + zy$

$xyz + xy + xz + yz$

$(xy + xyz) + xz + yz$

$= xy + xz + yz$ [as $xy + xyz = xy$]

X	Y	Z	XY	XZ	YZ	F	Minterm	Maxterm
0	0	0	0	0	0	0	-	M_0
0	0	1	0	0	0	0	-	M_1
0	1	0	0	0	0	0	-	M_2
0	1	1	0	0	1	1	m_3	
ϕ	0	0	0	0	0	0	-	M_4
1	0	1	0	1	1	1	m_5	
1	1	0	1	0	1	1	m_6	
1	1	1	1	1	1	1	m_7	

Canonical SOP $= \sum m(3,5,6,7)$

POS $= \prod M(0,1,2,4)$

Simplify using k-maps :-

(i) $F(w, x, y) = \sum m(1,4,5,6)$

minterms in binary:

1 = 001

4 = 100

5 = 101

6 = 110

k-map:

W \ XY	$\bar{X}\bar{Y}$	$\bar{X}Y$	XY	$X\bar{Y}$
$\bar{W}$	0	1 (1)	3	2
W	1 (4)	1 (5)	7	1 (6)

from the kaymap $\Rightarrow F = \bar{X}W + \bar{X}Y + W\bar{Y}$

(iii) $F(A,B,C,D,E) = \sum_m (0,2,3,4,5,6,7,11,15,16,18,19,23,$
$27,31)$

kmap $(E=0)$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1 (2)	1 (6)	1 (4)
$\bar{A}B$	8	10	14	12
AB	24	26	30	28
$A\bar{B}$	1 (16)	1 (18)	22	20

$E=1$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1 (3)	1 (7)	1 (5)
$\bar{A}B$	9	1 (11)	1 (15)	13
AB	25	1 (27)	1 (31)	29
$A\bar{B}$	17	1 (19)	1 (23)	21

$F = DE + \bar{A}\bar{B}C + \bar{B}C\bar{E}$

(iv) $F(A,B,C,D) = \prod M(0,1,2,3,4,10,11) \Rightarrow$ minterms :- 2, 5, 7, 6, 12, 13, 15, 14, 8, 9

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0:	1:	3:	2: 1
$\bar{A}B$	4:	5: 1	7: 1	6: 1
AB	12: 1	13: 1	15: 1	14: 1
$A\bar{B}$	8: 1	9: 1	11:	10:

$\bar{A}\bar{B} = 11$

$F = AB + AC' + A'BC + A'BD$

$= AB + AC' + A'B(C+D)$

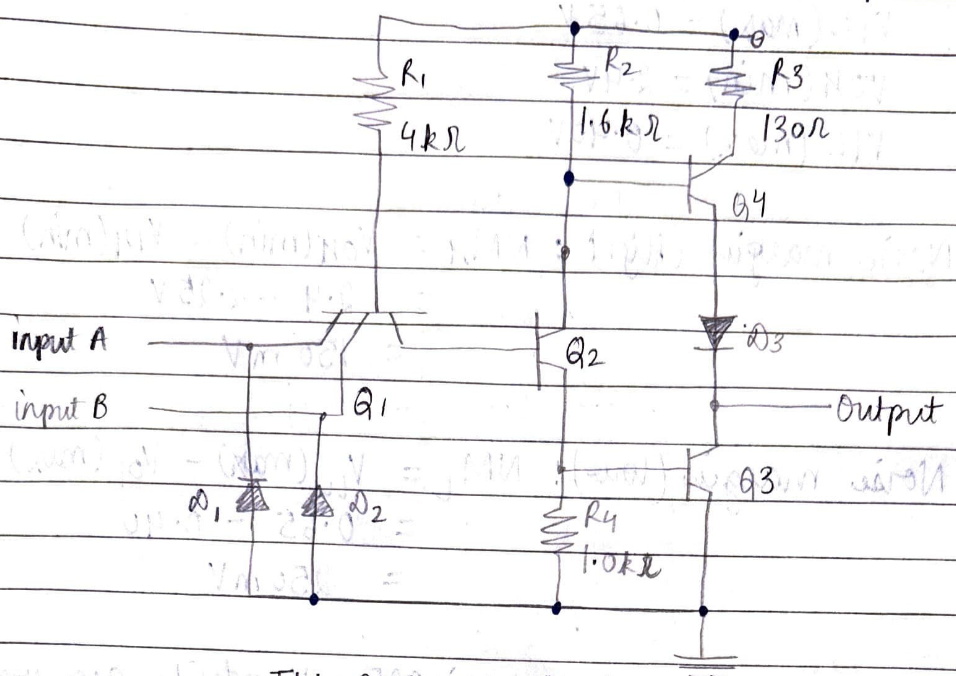
(iii) $A'B'C + B'C'D + BCD + ACD' + A'B'C + A'BC'D$

This expression yields minterms = 1, 2, 3, 5, 7, 9, 10, 14, 15

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0:	1: 1	3: 1	2: 1
$\bar{A}B$	4:	5: 1	7: 1	6:
AB	12:	13:	15: 1	14: 1
$A\bar{B}$	8:	9: 1	11:	10: 1

$F = A'D + AC + B'C + B'D$

Q4. (a) Draw TTL circuit with totem pole output



TTL AND gate circuit

verification:-

Case I: A=0 (low), B=0 (low)

⇒ Q1 (forward bias) ⇒ Q2 base LOW ⇒ Q2 off ⇒ Q4 off

pull out high through Q3

output = 1 (matches NAND)

case II: A=0, B=1 [and case III: A=1, B=0]

Q1 conducts emitter tied to A=0, B → pulled low ⇒

Q2 off ⇒ Q4 off ⇒ Q3 ON

output = 1 (matches NAND)

case IV: A=1, B=1

Q1 = high ⇒ Q1 off ⇒ B = high ⇒ Q2 = on ⇒ Q4 on

⇒ Q3 driven off

output = 0 (matches NAND)

(b) $V_{IH}(min) = 2.25\ V$
$V_{IL}(max) = 0.65\ V$
$V_{OH}(min) = 2.4V$
$V_{IL}\ (max) = 0.40V$

Noise margin (High) : $NM_H = V_{OH}(min) - V_{IH}(min)$
$= 2.4 - 2.25\ V$
$= 150\ mV$

Noise margin (low): $NM_L = V_{IL}\ (max) - V_{OL}\ (max)$
$= 0.65 - 0.40$
$= 250\ mV$

NM_H is 0.15 V and NM_L is 0.25 V which are very small value

∴ Hence, the high and low noise margins are not suitable for a circuit design. ~~as not~~